4_Struts2 _ Problem Assignments

Problem 1: Interceptors

Q1. Create an interceptor stack with timer & logger interceptor & use it in your action class
Q2. Identify the various interceptors currently provided in struts 2.x and what are they used for?
Q3. Write a simple program to print “Hello world” and using timer interceptor measure how long it took to execute an action method?

Problem 2: Custom Interceptors

Q4. Create custom interceptor for user authentication before login

Problem 3: Model Driven

Q5. Create a web project to create employee details using Model driven interceptors. On success the generated employee ID and Employee name should be displayed along with success message.
Employee details to be fetched are the following.

a) Employee Name
b) Location
c) Age
d) Gender
e) Date of Birth
f) PAN No.

Problem 4: UI Tags

Q6. Include a properties file for your Employee details scenario given in **Q5** and take all labels from the properties file.

Problem 5: Resource bundles

Q7. For the assignment provided in **Q6**

1) Use properties file for following validations in Employee details.
   a) Employee Name should be mandatory. (Field Error)
   b) Address field should be minimum 8 characters and maximum 50.(Field Error)
   c) Error messages in case of any errors. (Action Error)
2) Include a field Base Branch (drop-down) and designation (drop-down) in the Employee details page, and take the values for the field from a properties file.

Problem 6: Exception handling

For the assignment provided in **Q7**

- Create a global exception-mapping for all numeric exceptions that can happen while creating Employee details.
- Include a new field Salary of the employee, and should be calculated based on Designation.
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For example, salary for various designations should be within the limits mentioned below. Salary should be in Rupees with maximum 2 decimal places. Do the upper & lower limit validations for salary as mentioned below.
Include exception-mapping for salary computation.

Problem 7: Aware Interfaces

For the assignment provided in **Q7**

Q8. A new screen should be designed to capture the Band details of Employee. Name the screen as "Generate Band". This can be opened from a hyperlink in Employee details page.

- If Designation is Consultant and salary is between 2Lakh and 2.5Lakhs give band as 'A'
  If Designation is Consultant and salary is between 1.5Lakh and 2Lakhs give band as 'B'

- If Designation is Associate Consultant and salary is between 90000 and 1 lakh give band as 'A'
  If Designation is Associate Consultant and salary is between 80000 and 90000 give band as 'B'

- If Designation is Assistant Consultant and salary is between 70000 and 80000 give band as 'A'
  If Designation is Assistant Consultant and salary is between 60000 and 70000 give band as 'B'

- If Designation is IT Analyst and salary is between 40000 and 60000 give band as 'A'
  If Designation is IT Analyst and salary is between 35000 and 40000 give band as 'B'

- If Designation is Systems Engineer and salary is between 30000 and 35000 give band as 'A'
  If Designation is Systems Engineer and salary is between 25000 and 30000 give band as 'B'

Use application aware interface and display employee details as well in the Generate Band field.

Q9. Create a Login page for the employee application.

- Either Employee or Admin can log in to the application.
  Employee's Name & Designation/Role should be displayed on top of each page.

- Employee should be able to view his profile details. A separate screen should be made to view the Band for the employee.

- Admin should have the following screens.
  Create Employee details.
  View Employee details.
  Generate Band.